



2013 Half-Yearly Examination

FORM VI

MATHEMATICS 2 UNIT

Thursday 21st February 2013

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 85 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 75 Marks

- Questions 11–15 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 5 per boy
- Multiple choice answer sheet
- Candidature — 96 boys

Examiner
SG

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following is a primitive of $\sqrt{x}$?

- (A) $\frac{3}{2}x^{\frac{3}{2}}$
- (B) $\frac{2}{3}x^{\frac{3}{2}}$
- (C) $\frac{1}{2}x^{-\frac{1}{2}}$
- (D) $-\frac{1}{2}x^{-\frac{1}{2}}$

QUESTION TWO

What is the value of the definite integral $\int_{-1}^2 x^2 dx$?

- (A) 3
- (B) -3
- (C) $\frac{7}{3}$
- (D) $-\frac{7}{3}$

QUESTION THREE

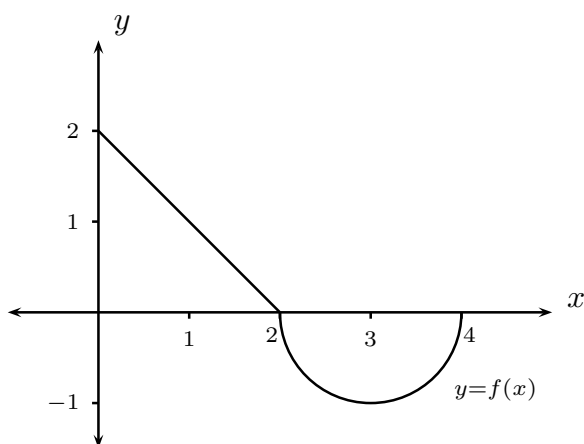
A point $P(x, y)$ moves so that it is always equidistant from the points $A(0, 0)$ and $B(5, 5)$. Which of the following best describes the locus of P ?

- (A) A line
- (B) A circle
- (C) A parabola
- (D) A hyperbola

QUESTION FOUR

The graph of $y = f(x)$ is shown below. It consists of a straight line section and a semicircle.

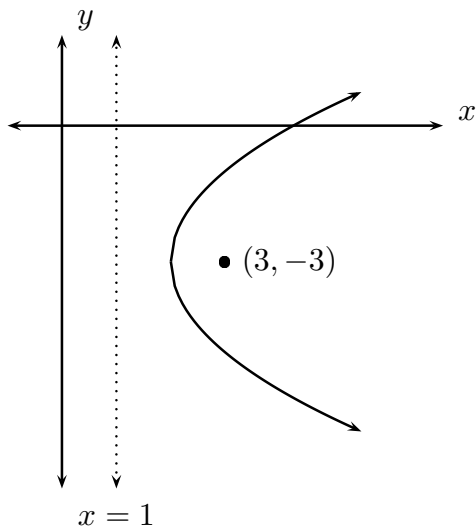
What is the value of the definite integral $\int_0^4 f(x) dx$?



- (A) $2 + \pi$
- (B) $2 - \pi$
- (C) $2 + \frac{\pi}{2}$
- (D) $2 - \frac{\pi}{2}$

QUESTION FIVE

A parabola has its focus at $(3, -3)$ and directrix at $x = 1$. What is the equation of this parabola?



- (A) $(y + 3) = 4(x - 2)^2$
- (B) $(y + 3)^2 = 4(x - 2)$
- (C) $(y - 3)^2 = 4(x + 2)$
- (D) $(y - 3) = 4(x + 2)^2$

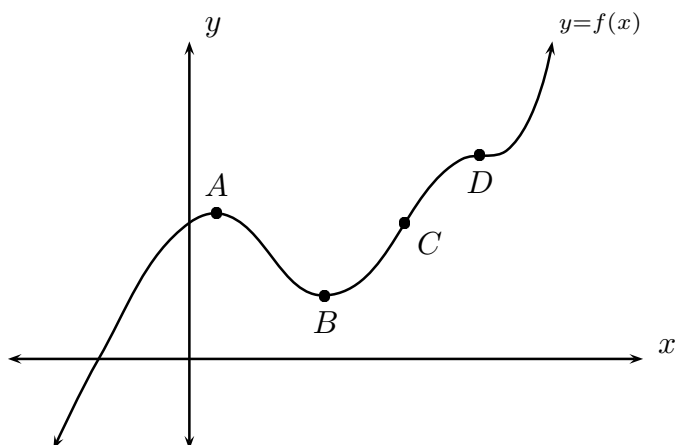
QUESTION SIX

The graph of $y = e^x$ is translated two units to the right. Which of the following represents the new function?

- (A) $y = e^{x+2}$
- (B) $y = e^{x-2}$
- (C) $y = e^x + 2$
- (D) $y = e^x - 2$

QUESTION SEVEN

For $y = f(x)$ graphed below, which of the labelled points satisfies $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$?



- (A) A
- (B) B
- (C) C
- (D) D

QUESTION EIGHT

Which of the following is not true of the function $f(x) = e^x + 3$?

- (A) The first derivative is e^x .
- (B) The function is always increasing.
- (C) The function has its y -intercept at $(0, 1)$.
- (D) A primitive of the function is $e^x + 3x$.

QUESTION NINE

The function $f(x) = (x - 2)^2(x + 7)$ has the following first and second derivatives:

$$f'(x) = 3(x - 2)(x + 4)$$

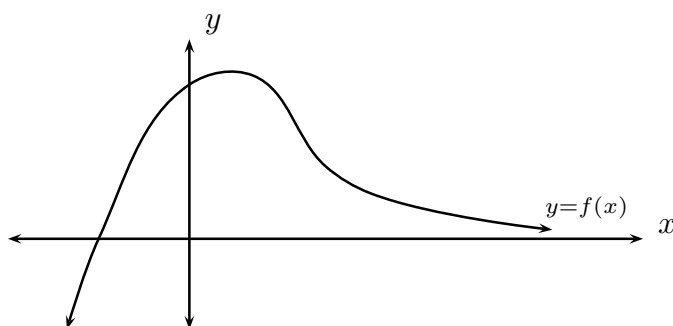
$$f''(x) = 6(x + 1)$$

Which of the following statements is false?

- (A) $f(x)$ has stationary points at $x = 2$ and $x = -4$.
- (B) $f(x)$ is concave up for $x < -1$.
- (C) $f(x)$ has x -intercepts at $x = 2$ and $x = -7$.
- (D) $f(x)$ is a cubic function.

QUESTION TEN

A function $y = f(x)$ is graphed below. The x -axis is an asymptote for the function.



Which of the following statements is true?

- (A) $f(x)$ has a global minimum at $y = 0$.
- (B) $f(x)$ has an asymptote at $x = 0$.
- (C) $f(x)$ has a single point of inflexion.
- (D) $f(x)$ has two stationary points.

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.

Marks

- | | |
|--|--|
| (a) Calculate $\frac{3e^2}{5}$, correct to three decimal places. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (b) Simplify $\frac{(e^x)^3}{e^{2x}}$. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">2</div> |
| (c) (i) Sketch the locus of a point P which moves so that it is always at a fixed distance of two units below the x -axis. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (ii) Write down the equation of the locus. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (d) A parabola has equation $x^2 = -4y$. | |
| (i) Write down the coordinates of the vertex. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (ii) Write down the coordinates of the focus. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (iii) Write down the equation of the directrix. | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (iv) Sketch the parabola, showing all the features found in (i) to (iii). | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (e) Differentiate the following with respect to x : | |
| (i) $5x^3 - 3x^2 + 9$ | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (ii) $4e^{5x}$ | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (iii) $(2 - 3x)^4$ | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">2</div> |
| (f) Find a primitive for each of the following: | |
| (i) $x^2 - 2$ | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |
| (ii) $x^{-\frac{1}{2}}$ | <div style="border: 1px solid black; padding: 2px 5px; display: inline-block;">1</div> |

Exam continues overleaf ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) Expand and simplify $(e^x + e^{-x})^2$.

2

(b) Evaluate the following definite integrals:

(i) $\int_{-2}^2 (3x^2 + 2x) dx$

2

(ii) $\int_1^3 \frac{2}{x^2} dx$

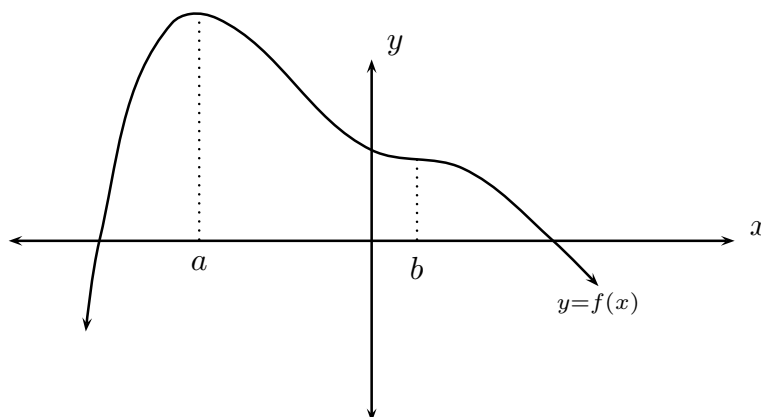
2

(c) Given $f'(x) = e^x + 2x$, find $f(x)$ if $f(0) = 0$.

2

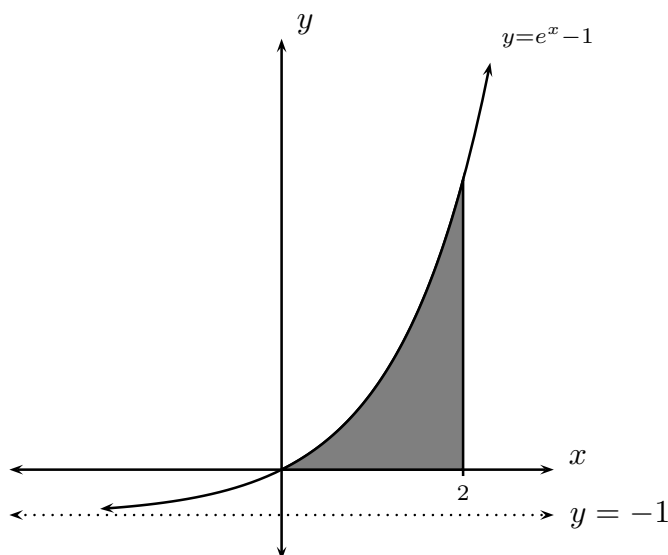
(d) The following diagram shows the graph of $y = f(x)$. A maximum occurs when $x = a$ and a stationary point of inflexion when $x = b$. Sketch a possible graph of $f'(x)$.

2



(e) Find the area of the region bounded by the curve $y = e^x - 1$, the x -axis and the line $x = 2$, as shown in the diagram below. Express your answer in terms of e .

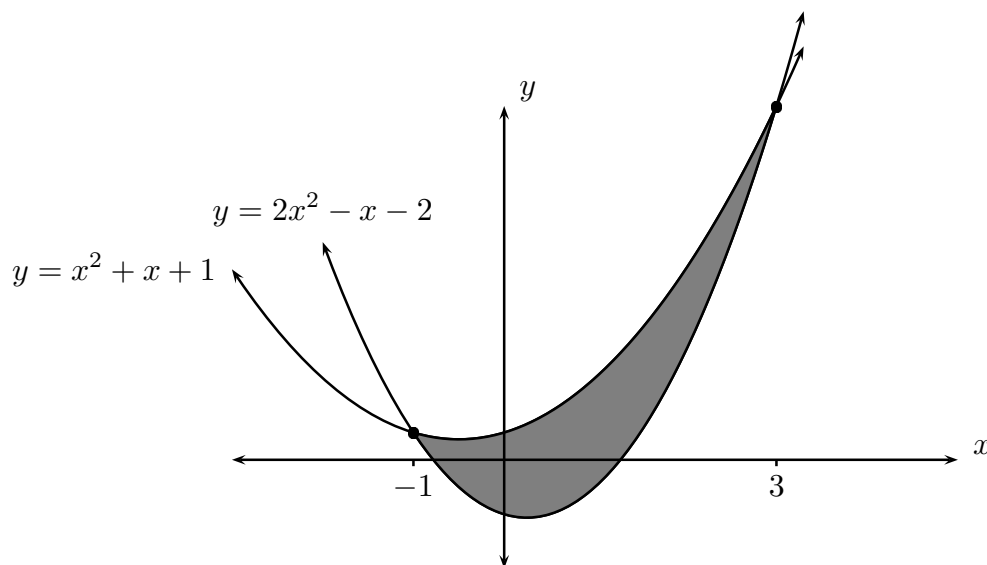
2



QUESTION TWELVE (Continued)

- (f) The curves $y = x^2 + x + 1$ and $y = 2x^2 - x - 2$ meet at two points whose x -coordinates are $x = -1$ and $x = 3$.

3



Find the area of the shaded region enclosed between the two curves, as shown in the diagram above.

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) Consider the curve $y = 3x^3 - 9x + 3$.

(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

2

- (ii) Find all the stationary points of the curve.

2

- (iii) Determine the nature of each stationary point.

2

- (iv) Sketch the curve, clearly indicating all the stationary points and the y -intercept.

2

Note, you are not required to find the x -intercepts or any points of inflexion.

(b) Evaluate $\int_{-2}^2 e^{2x+1} dx$.

2

- (c) Using Simpson's rule with five function values, estimate $\int_1^5 \frac{1}{x} dx$ correct to three decimal places.

2

- (d) Find the equation of the tangent to the curve $y = e^{3x}$ at the point where $x = 0$.

3

Exam continues overleaf ...

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

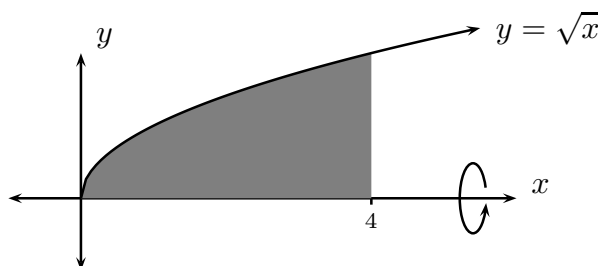
- (a) The cost,
- C
- dollars, of running a vehicle at an average speed of
- v
- km/h is given by

3

$$C = \frac{2}{5}v + 2000v^{-1}, \quad \text{where } v > 0.$$

For what average speed will the cost be minimised?

- (b) The region bounded by the curve
- $y = \sqrt{x}$
- , the
- x
- axis and the line
- $x = 4$
- , is shown below.

2Find the volume of the solid generated when this region is rotated about the x -axis.

- (c) A function
- $f(x)$
- has second derivative
- $f''(x) = 20(x-1)^2(x-4)$
- . Show that
- $f(x)$
- has only one point of inflexion.

2

- (d) Differentiate the following, leaving your answers in simplest form.

(i) $y = (2x-1)e^x$

2

(ii) $y = \frac{e^x}{2x+3}$

3

- (e) (i) Differentiate
- $y = e^{x^3}$
- .

1

(ii) Hence evaluate the definite integral $\int_0^1 3x^2 e^{x^3} dx$.

2

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.

Marks

(a) Consider the function $f(x) = \frac{1}{(x+4)^2}$.

(i) Find $f''(x)$.

2

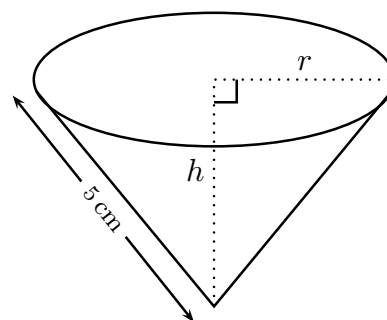
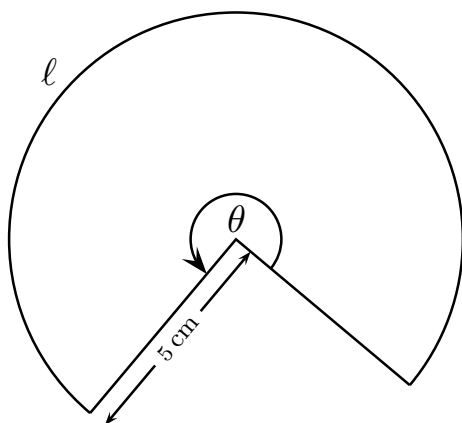
(ii) Explain why $y = f(x)$ is concave up for all real x except $x = -4$.

1

(b) Show that the function $y = xe^{-2x}$ satisfies the equation $y'' + 4y' + 4y = 0$.

3

(c) A sector with radius 5 cm, arc length ℓ cm and angle θ degrees at its centre is bent to form a cone, as shown in the diagram below. The resultant cone has base radius r cm and height h cm.



(i) Show that $\ell = \frac{\pi\theta}{36}$.

1

(ii) Hence show that $r = \frac{\theta}{72}$.

1

(iii) Show that $h = \sqrt{25 - \left(\frac{\theta}{72}\right)^2}$.

1

(iv) Construct an equation for the volume of the cone $V \text{ cm}^3$ as a function of θ only.

2

(v) Find, to the nearest degree, the value of θ for which the volume of the cone is maximised.

4

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2013
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FORM VI
MATHEMATICS 2 UNIT
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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

FORM VI 2 UNIT HALF-YEARLY SOLUTIONS, 2013

Q1. B

Q2. A

Q3. A

Q4. D

Q5. B

Q6. B

Q7. A

Q8. C

Q9. B

Q10. C

Q11.

(a) $\frac{3e^2}{5} \div 4.433$ ✓

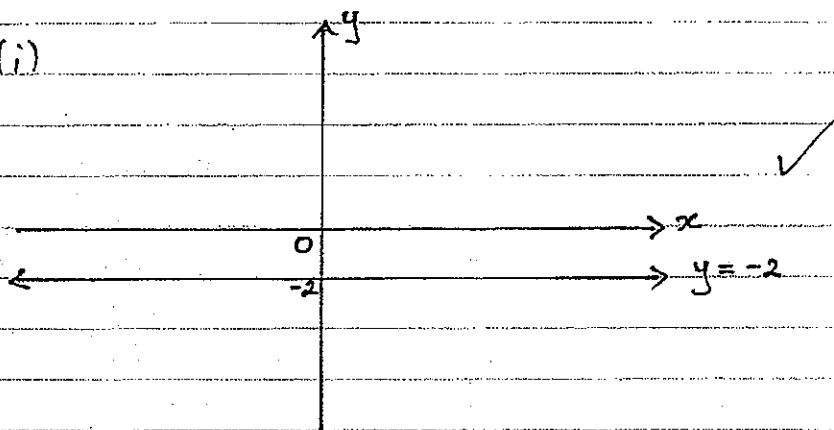
①

(b) $\frac{(e^x)^3}{e^{2x}} = \frac{e^{3x}}{e^{2x}}$ ✓

$= e^x$ ✓

②

(c) (i)



(ii) $y = -2$ ✓

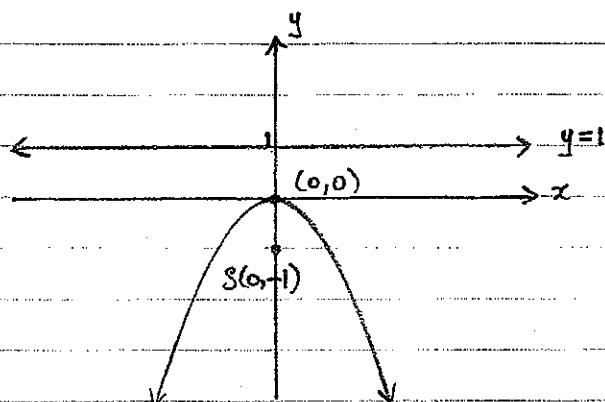
②

(d) (i) $(0, 0)$ ✓

(ii) $4a = 4 \rightarrow a = 1$ Parabola concave down, so

focus at $S(0, -1)$. ✓

(iii) Directrix one unit above vertex: $y = 1$ ✓



Directrix, vertex,
focus needed
on sketch for
mark.

④

$$(e) \quad (i) \quad \frac{d}{dx} (5x^3 - 3x^2 + 9) = 15x^2 - 6x \quad \checkmark$$

$$(ii) \quad \frac{d}{dx} 4e^{5x} = 20e^{5x} \quad \checkmark$$

$$(iii) \quad \frac{d}{dx} (2-3x)^4 = 4(2-3x)^3 (-3) \quad \checkmark$$

$$= -12(2-3x)^3 \quad \checkmark$$

④

$$(f) \quad (i) \quad x^3 - 2x + C \quad (\text{any } C, \text{ constant}) \quad \checkmark$$

$$(ii) \quad \int x^{-1/2} = 2x^{1/2} + C \quad (\text{any } C, \text{ constant}) \quad \checkmark$$

②

15

Q12

$$(a) \quad (e^x + e^{-x})^2 = e^{2x} + 2e^x e^{-x} + e^{-2x} \quad \checkmark$$

$$= e^{2x} + e^{-2x} + 2 \quad \checkmark$$

②

$$(b) \quad (i) \quad \int_{-2}^2 (3x^2 + 2x) dx = \left. x^3 + x^2 \right|_{-2}^2 \quad \checkmark$$

$$= (2^3 + 2^2) - ((-2)^3 + (-2)^2)$$

$$= 16 \quad \checkmark$$

$$\begin{aligned}
 \text{(ii)} \quad \int_1^3 \frac{2}{x^2} dx &= \int_1^3 2x^{-2} dx \\
 &= \left[-2x^{-1} \right]_1^3 \\
 &= -2 \left(\frac{1}{3} - 1 \right)
 \end{aligned}$$

$$= \frac{4}{3} \quad \text{or} \quad 1\frac{1}{3} \quad \text{✓} \quad \textcircled{4}$$

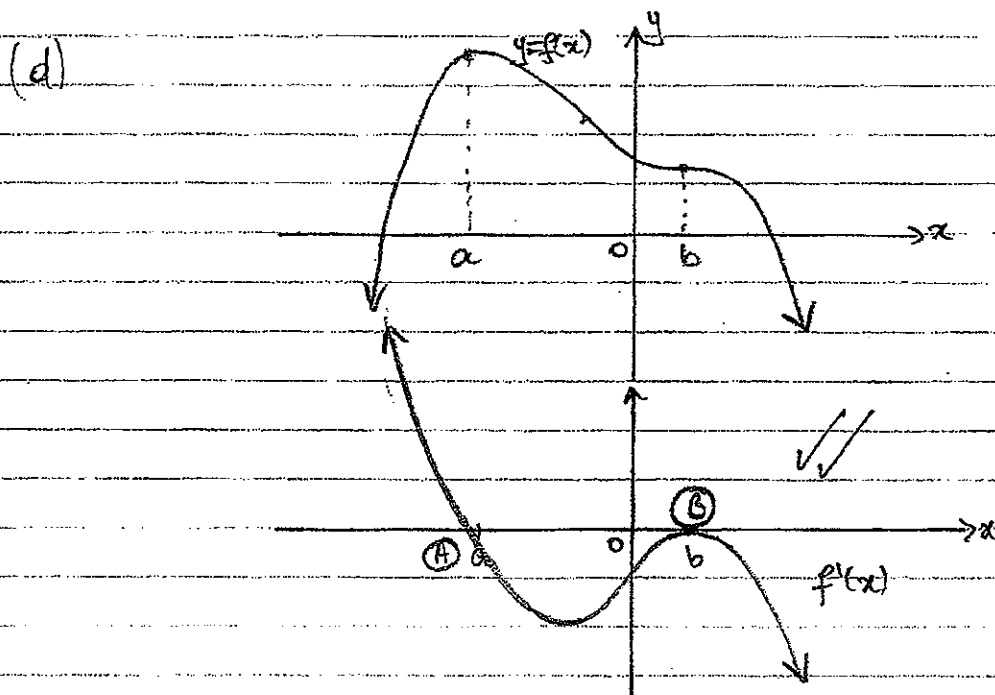
$$\text{(c)} \quad f'(x) = e^x + 2x \rightarrow f(x) = e^x + x^2 + C \quad \text{✓}$$

$$\begin{aligned}
 \text{Now,} \quad f(0) &= e^0 + 0 + C \\
 &= 0
 \end{aligned}$$

$$\text{i.e.} \quad 1 + C = 0$$

$$\therefore C = -1$$

$$\text{Hence} \quad f(x) = e^x + x^2 - 1 \quad \text{✓} \quad \textcircled{2}$$



1 For any of (A) or (B)
2 for both (A) + (B)

②

$$(e) \int_0^2 (e^x - 1) dx = \left[e^x - x \right]_0^2 \quad \checkmark$$

$$= (e^2 - 2) - (e^0 - 0)$$

$$= e^2 - 3 \quad \checkmark \quad (2)$$

$$(f) \int_{-1}^3 (x^2 + x + 1) - (2x^2 - x - 2) dx = \int_{-1}^3 -x^2 + 2x + 3 dx \quad \checkmark$$

$$= \left[-\frac{x^3}{3} + x^2 + 3x \right]_{-1}^3 \quad \checkmark$$

$$= \left(\left(-\frac{3^3}{3} + 3^2 + 3(3) \right) - \left(-\frac{(-1)^3}{3} + (-1)^2 + 3(-1) \right) \right)$$

$$= \frac{32}{3} \quad \checkmark \quad (3)$$

15

Q13

(a) $y = 3x^3 - 9x + 3$

(i) $\frac{dy}{dx} = 9x^2 - 9$ ✓, $\frac{d^2y}{dx^2} = 18x$ ✓
 $= 9(x^2 - 1)$ ✓ either

(ii) Need all x : $\frac{dy}{dx} = 0$

Hence $9(x^2 - 1) = 0 \rightarrow x = \pm 1$ ✓

Corresponding y -coordinates: $y(1) = 3 - 9 + 3 = -3$

$y(-1) = -3 + 9 + 3 = 9$

Stationary points: $(1, -3)$ and $(-1, 9)$ ✓

(iii) Nature: $\frac{d^2y}{dx^2} = 18x$, so for:

$x = 1$, $\frac{d^2y}{dx^2} = 18 > 0 \rightarrow$ concave up, local min

$x = -1$, $\frac{d^2y}{dx^2} = -18 < 0 \rightarrow$ concave down, local max. ✓

Hence $(1, -3)$ local min

$(-1, 9)$ local max. ✓

OR use table of gradients:

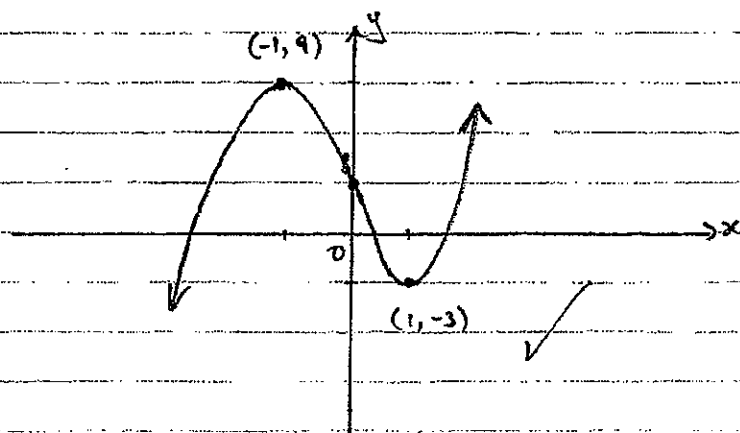
x	-2	-1	0	1	2
$\frac{dy}{dx}$	27	0	-9	0	27

✓

/ \ / \ /

implies $(1, -3)$ local min. ✓
 $(-1, 9)$ local max. ✓

(iv)



y-intercept at (0, 3)

Full marks (2) only
for both
y-intercept
&

stationary points.
(8)

$$(b) \int_{-2}^2 e^{2x+1} dx = \left[\frac{1}{2} e^{2x+1} \right]_{-2}^2$$

$$= \frac{1}{2} (e^5 - e^{-3})$$

OR $= \frac{e^8 - 1}{2e^3}$

$$(c) \int_1^5 \frac{1}{x} dx \approx \frac{3-1}{6} \left[1 + 4 \times \frac{1}{2} + \frac{1}{3} \right] + \frac{5-3}{6} \left[\frac{1}{2} + 4 \times \frac{1}{4} + \frac{1}{5} \right]$$

x	1	2	3	4	5
f(x)	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$

$$= \frac{73}{45} \approx 1.622 \text{ (3 d.p.)}$$

$$(d) y = e^{3x} \rightarrow y' = 3e^{3x}$$

At $x=0$, $y'(0) = 3$

Also, $y(0) = e^0 = 1$

$\therefore$ line is: $y - 1 = 3(x - 0)$

$$y = 3x + 1$$

OR $3x - y + 1 = 0$

Q14. (a) $C = \frac{2v}{5} + 2000v^{-1}$, $v > 0$.

$$\frac{dC}{dv} = \frac{2}{5} - \frac{2000}{v^2}$$

Extrema for v such that $\frac{dC}{dv} = 0$

Hence,

$$\frac{2}{5} - \frac{2000}{v^2} = 0$$

$$\frac{2000}{v^2} = \frac{2}{5}$$

$$\begin{aligned} v^2 &= 5000 \\ \therefore v &= \pm \sqrt{5000} \\ &= \pm 50\sqrt{2} \end{aligned}$$

$$\begin{aligned} \rightarrow v &= 50\sqrt{2} \text{ km/h } (\because v > 0) \\ \text{OR } v &= 70.7 \text{ km/h}^{-1} \end{aligned}$$

Any

Test nature:

$$\frac{d^2C}{dv^2} = -2000 \times (-2) v^{-3}$$

> 0 for all $v > 0$, so global min. where $v = 50\sqrt{2} \text{ km/h}$.

OR

$$\frac{d^2C}{dv^2} \text{ at } v = 50\sqrt{2} : \frac{-2000 \times (-2)}{(50\sqrt{2})^3} > 0 \text{ accepted.}$$

$\therefore$ local minimum where $v = 50\sqrt{2} \text{ km/h}$. (3)

Accept approximations too: $50\sqrt{2} \text{ km/h} \approx 70.71 \text{ km/h}$.

OR table of gradients used to test nature of stationary point for one mark.

(b) $y = \sqrt{x}$

Rotation about x-axis: $V = \pi \int_0^4 (\sqrt{x})^2 dx$

$$= \pi \int_0^4 x dx \quad \checkmark$$

$$= \pi \left[\frac{x^2}{2} \right]_0^4$$

$$= \pi \left[\frac{4^2}{2} - 0 \right]$$

$$= 8\pi \quad \checkmark \quad (2)$$

(c) $f''(x) = 20(x-1)^2(x-4)$

$f(x)$ will have potential points of inflexion for $f''(x) = 0$.

Hence, $20(x-1)^2(x-4) = 0$

implies $x = 1$ or $x = 4$ $\checkmark$

Test concavity change:

x	0	1	2	4	5
$f''(x)$	< 0	.	< 0	.	> 0
concavity	$\cap$	.	$\cap$	.	$\cup$

Change in concavity only where $x = 4$.

$\therefore f(x)$ has only one point of inflexion $\checkmark$

(2)

$$(d) (i) \frac{dy}{dx} = e^x(2) + (2x-1)e^x \quad \checkmark$$

$$= e^x(2+2x-1)$$

$$= (2x+1)e^x \quad \checkmark$$

$$(ii) \frac{dy}{dx} = \frac{(2x+3)e^x - e^x(2)}{(2x+3)^2} \quad \checkmark$$

$$= \frac{e^x(2x+1)}{(2x+3)^2} \quad \checkmark$$

④

$$(e) (i) y = e^{x^3} \rightarrow \frac{dy}{dx} = 3x^2 e^{x^3} \quad \checkmark \checkmark$$

OR $y = e^{x^3}$. Let $u = x^3$. Then $y = e^u$ ✓

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= e^u \cdot 3x^2$$

$$= 3x^2 e^{x^3} \quad \checkmark$$

$$(ii) \therefore \int_0^1 3x^2 e^{x^3} dx = e^{x^3} \Big|_0^1 \quad \checkmark$$

$$= e^1 - e^0$$

$$= e - 1 \quad \checkmark$$

④

Q15

$$(a) \quad f(x) = \frac{1}{(x+4)^2}, \quad x \neq -4.$$

$$= (x+4)^{-2}$$

$$\therefore f'(x) = -2(x+4)^{-3}$$

$$\& \therefore f''(x) = \frac{6}{(x+4)^4} \quad \checkmark$$

Now, $(x+4)^4 > 0$ for all real x , $x \neq -4$. Hence

$$\frac{6}{(x+4)^4} > 0 \quad \text{for all real } x, x \neq -4. \quad \checkmark$$

Therefore $f''(x) > 0$ on this domain,

so $f(x)$ is concave up for all x real,

$x \neq -4$. $\checkmark$

(3)

$$(b) \quad y = xe^{-2x} \rightarrow y' = x \cdot (-2)e^{-2x} + e^{-2x} \cdot 1$$

$$= -2xe^{-2x} + e^{-2x} \quad \checkmark$$

$$\& \quad y'' = -2x \cdot (-2)e^{-2x} + e^{-2x} \cdot (-2) + (-2)e^{-2x}$$

$$= 4xe^{-2x} - 4e^{-2x} \quad \checkmark$$

$$\text{Now, } y'' + 4y' + 4y = 4xe^{-2x} - 4e^{-2x} + 4(-2xe^{-2x} + e^{-2x})$$

$$+ 4(xe^{-2x})$$

$$= \cancel{4xe^{-2x}} - \cancel{4e^{-2x}} - \cancel{8xe^{-2x}} + \cancel{4e^{-2x}} + 4xe^{-2x}$$

$$= 0 \quad \checkmark$$

(3)

$$(c)(i) \frac{l}{2\pi(5)} = \frac{\theta}{360} \rightarrow l = \frac{10\pi\theta}{360} \quad \text{Accept: } l = r\theta \times \frac{\pi}{180}$$

$$\text{i.e. } l = \frac{\pi\theta}{36}$$

$$\begin{aligned} &= 5\theta \times \frac{\pi}{180} \\ &= \frac{\pi\theta}{36} \end{aligned}$$

(ii) Circumference of base of cone is equal to arc length of sector:

$$\therefore l = 2\pi r$$

Since $l = \frac{\pi\theta}{36}$, it follows,

$$\frac{\pi\theta}{36} = 2\pi r$$

$$\therefore r = \frac{\theta}{72}$$

(iii) By Pythagoras' theorem,

$$5^2 = r^2 + h^2$$

$$\text{hence } h = \sqrt{25 - r^2} \quad \checkmark$$

$$= \sqrt{25 - \left(\frac{\theta}{72}\right)^2} \quad \checkmark$$

$$(iv) V = \frac{1}{3}\pi r^2 h \quad \checkmark$$

$$\therefore V = \frac{1}{3}\pi \left(\frac{\theta}{72}\right)^2 \sqrt{25 - \left(\frac{\theta}{72}\right)^2} \quad \text{or equivalent} \quad \checkmark$$

$$(v) \frac{dV}{d\theta} = \frac{1}{3} \cdot \frac{\pi}{72^2} \left[2\theta \sqrt{25 - \frac{\theta^2}{72^2}} + \theta^2 \cdot \frac{1}{2\sqrt{25 - \frac{\theta^2}{72^2}}} \cdot \frac{-2\theta}{72^2} \right] \checkmark$$

Set $\frac{dV}{d\theta} = 0$ & solve for θ : implies,

$$2\theta \sqrt{25 - \frac{\theta^2}{72^2}} - \frac{\theta^3}{72^2 \sqrt{25 - \frac{\theta^2}{72^2}}} = 0$$

$$\theta \left(2 \sqrt{25 - \frac{\theta^2}{72^2}} - \frac{\theta^2}{72^2 \sqrt{25 - \frac{\theta^2}{72^2}}} \right) = 0$$

$\therefore \theta = 0$ (reject $\because$ cone volume clearly not maximised for $\theta = 0$).

or

$$2\sqrt{25 - \frac{\theta^2}{72^2}} = \frac{\theta^2}{72^2 \sqrt{25 - \frac{\theta^2}{72^2}}}$$

i.e.

$$2 \cdot 72^2 (25 - \frac{\theta^2}{72^2}) = \theta^2$$

$$2 \cdot 25 \cdot 72^2 - 2\theta^2 = \theta^2$$

$$\theta^2 = 86400$$

so

$$\theta = 120\sqrt{6} \quad (\text{+ve soln only})$$

$$= 293.938\dots$$

$$\approx 294^\circ \quad \checkmark$$

Test nature of stationary point:

θ	290°	$120\sqrt{6}^\circ$	300°
$\frac{dV}{d\theta}$	≈ 0.026	0	≈ -0.046
slope	/	—	\

$\therefore$ Volume is at a maximum when $\theta = 120\sqrt{6}^\circ$
or $\theta \approx 294^\circ$ ✓

Note: incorrect response to part (iv) but correct differentiation received full marks.

⑨

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